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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS2E48 - SOFTWARE TESTING

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1 Which of the following is a dynamic testing technique?**
a) Code review b) Unit testing c) Walkthrough d) Inspection
2. **CO1 The first phase of the Testing Life Cycle (TLC) is _____.**
a) Test Execution b) Test Planning c) Test Closure d) Defect Reporting
3. **CO3 Selenium is primarily used for _____.**
a) Database testing b) Web application automation testing
c) Load testing d) Requirement gathering
4. **CO1 Test scripts are used to _____.**
a) Deploy software b) Execute test cases manually or automatically
c) Write functional specifications d) Design UI
5. **CO4 Which of the following is a non-functional testing example?**
a) Checking page load time under heavy traffic
b) Testing login with invalid credentials
c) Validating form fields d) Checking SQL queries

PART - B

(5x 3 = 15 Marks)

6. **CO1** Generate test data for a bank account module.
7. **CO3** Apply any one testing technique to find defects in a profile update module.

8. **CO2** Write a simple test script for login validation.
9. **CO4** Compare load testing with stress testing.
10. **CO4** Discuss the five different views of software quality.

PART - C

(5 x 10 = 50 Marks)

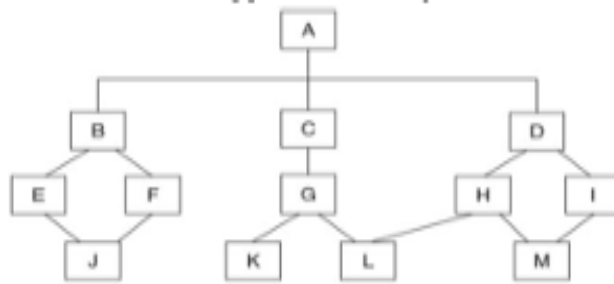
Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) Illustrate how test cases are derived from requirements with suitable example. (5)
CO1 ii) Discuss the impact of software testing on every phase of the SDLC, providing real-world examples for each phase. (5)
12. **CO2** i) Compare and contrast manual and automated test scripts, highlighting their advantages, limitations, and use cases. (5)
CO2 ii) Explain the difference between manual and automated testing techniques using a real-time application scenario. (5)
13. **CO3** i) Discuss how functional testing validates business requirements with an example of an online shopping portal. (5)
CO3 ii) Evaluate the effectiveness of white box testing in detecting logic-related defects compared to black box testing. (5)
14. **CO3** Develop a complete security testing strategy for a healthcare management system, including authentication, authorization, and data privacy aspects. (10)
15. **CO3** Design a tested database framework for an e-learning platform, including data validation, migration testing, and referential integrity verification. (10)

16. **CO4** Using the module hierarchy given in Figure, show the orders of module integration for the top-down and bottom-up integration approaches. Estimate the number of stubs and drivers needed for each approach. Specify the integration testing activities that can be done in parallel, assuming you have three SIT engineers. Based on the resource needs and the ability to carry out concurrent SIT activities, which approach would you select for this system and why? (10)



17. **CO4** For the current software project, you are working on, answer the following questions: i. List the types of software reviews (such as peer review, walkthrough, inspection, etc.) applied in your project. ii. Analyze why these specific reporting and control measures are most suitable for ensuring project transparency and timely issue resolution. (10)
